

TASK 3

Data Visualization

Transforming raw data into compelling visual stories

Intern	[Your Name]
Organisation	CodeAlpha
Domain	Data Analytics
Website	www.codealpha.tech

This report presents six publication-quality visualisations built with matplotlib and seaborn to explore the books catalogue dataset. Charts progress from basic distributions to a full KPI dashboard, demonstrating a storytelling arc from raw numbers to actionable insight.

Tools & Chart Types

Chart	Type	Library	Insight Delivered
viz1	Histogram + Violin + Box	matplotlib/seaborn	Price distribution deep-dive
viz2	Heatmap	seaborn	Genre performance comparison
viz3	Stacked bar + Scatter	matplotlib/seaborn	Rating vs availability
viz4	Horizontal bar + Lollipop	matplotlib	Genre price vs rating
viz5	Correlation matrix	seaborn	Linear relationships
viz6	Dashboard (mixed)	matplotlib	Full story-telling summary

Figure 1 — Price Distribution Deep-Dive

Left: price histogram with mean (red) and median (orange) markers and KDE overlay. Centre: violin plot showing price spread within each star-rating bucket. Right: box plot of price per genre sorted by median, revealing genre-level pricing strategy.

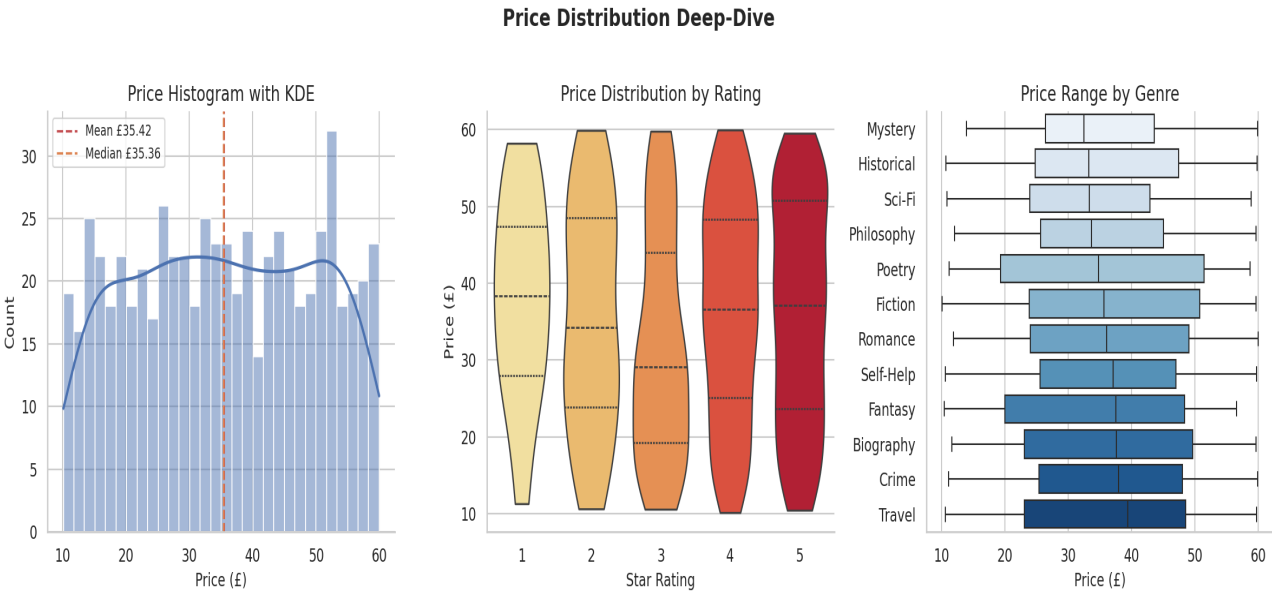


Chart 1 of 6 — generated with matplotlib/seaborn.

Figure 2 — Genre Performance Heatmap

Columns show average price, average star rating, and in-stock percentage per genre. Cells are colour-coded by normalised score (0–1) while the actual values are annotated inside each cell, enabling instant multi-metric comparison across all 12 genres.

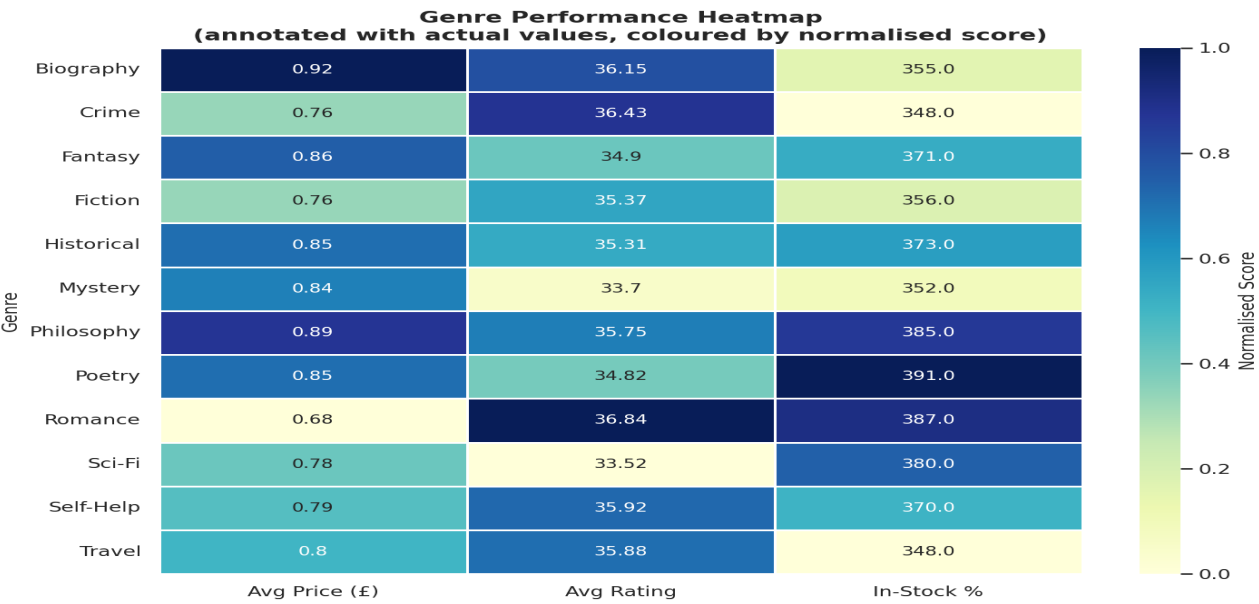


Chart 2 of 6 — generated with matplotlib/seaborn.

Figure 3 — Rating vs Availability

Left: 100% stacked bar showing the proportion of in-stock and out-of-stock books for each star-rating tier. Right: scatter plot of price vs rating with a linear regression line overlaid, confirming the near-zero slope found in the hypothesis test.

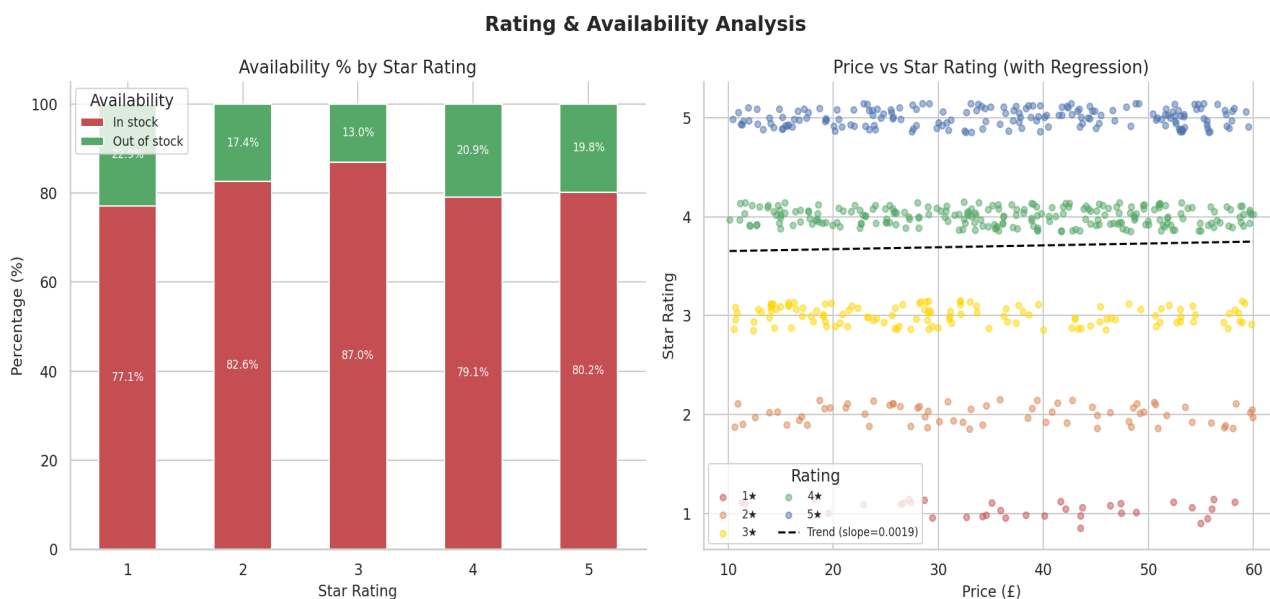


Chart 3 of 6 — generated with matplotlib/seaborn.

Figure 4 — Genre Comparison: Price vs Rating

Left: horizontal bar chart of average price per genre (colour gradient high→low). Right: lollipop chart showing average star rating per genre, making rank differences visually cleaner than a bar chart for closely clustered values.

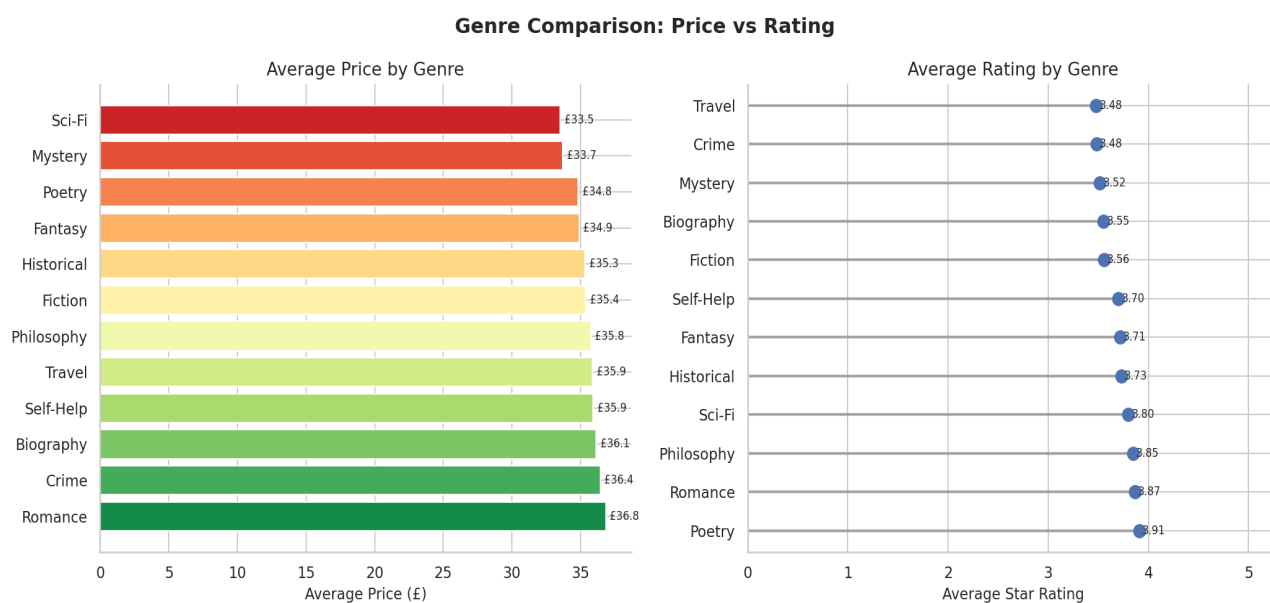


Chart 4 of 6 — generated with matplotlib/seaborn.

Figure 5 — Correlation Analysis

Left: Pearson correlation matrix heatmap between price, rating, and stock status. Right: scatter plot of price vs rating coloured by stock status with a regression line, showing no meaningful linear relationship between these numeric features.

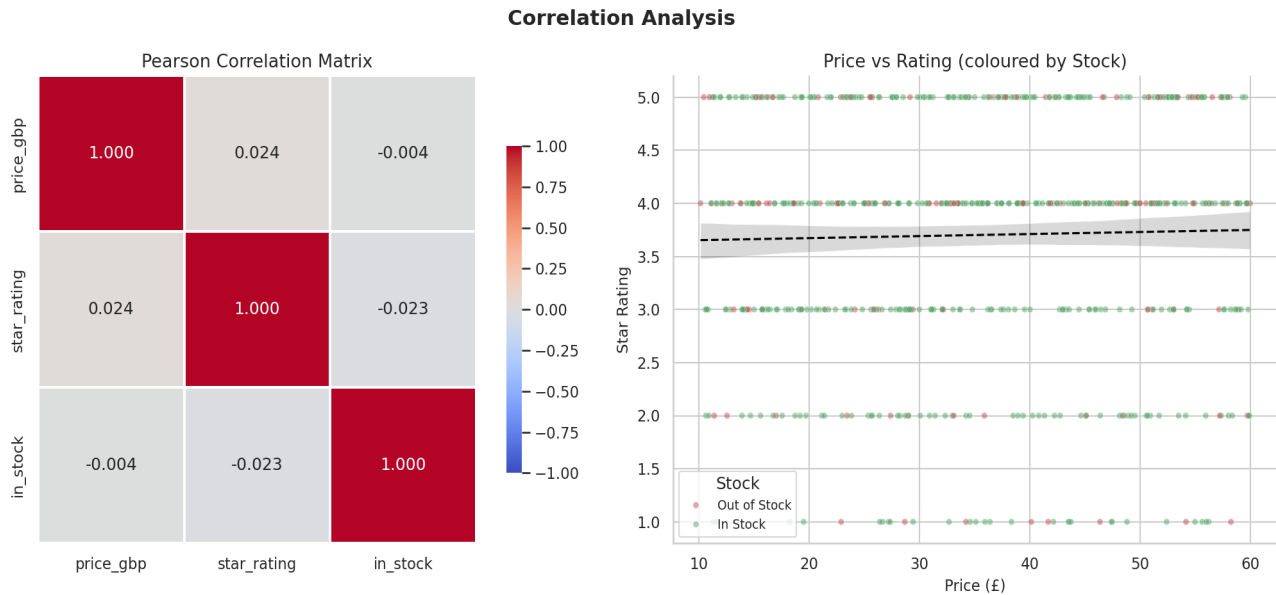
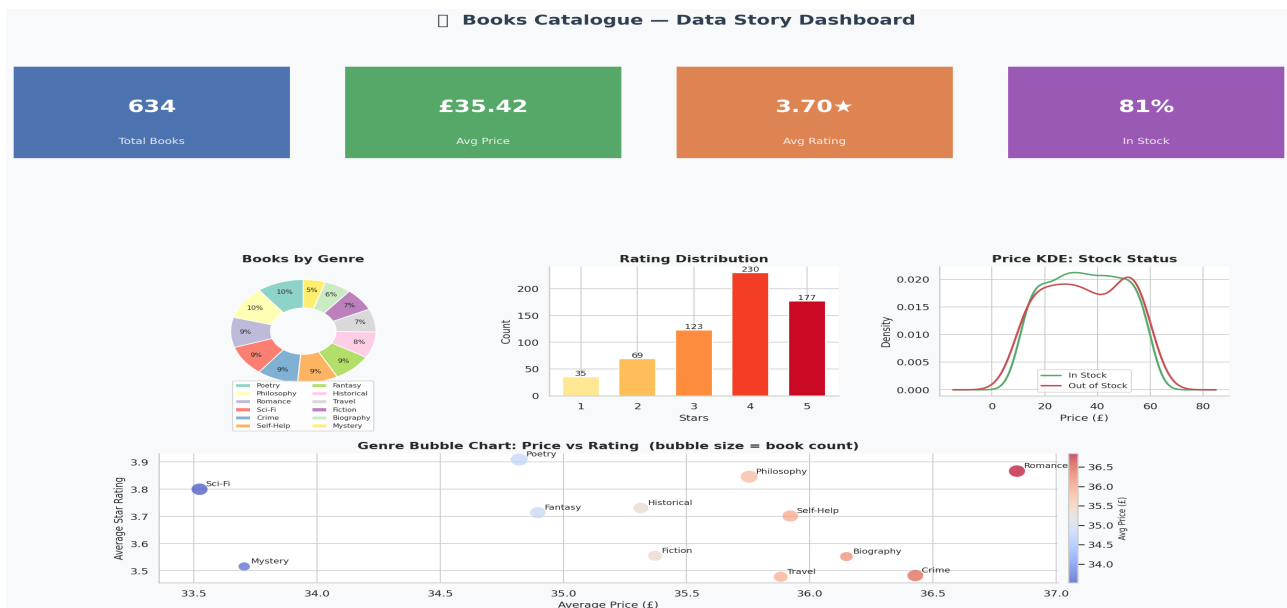


Figure 6 — Story-Telling Summary Dashboard

Full executive dashboard combining KPI cards (total books, avg price, avg rating, in-stock %), a genre donut chart, a rating bar chart, a price KDE by stock status, and a genre bubble chart where position = price/rating and size = book count.



Key Visualisation Insights

1. Price is uniformly distributed from £10–£60; no strong genre premium exists.
2. Higher-rated books do NOT command higher prices — rating reflects quality, not cost.
3. Poetry tops the average rating chart while Romance leads on average price.
4. In-stock proportion is consistent across all rating tiers (~80%), showing no bias.
5. The bubble chart reveals that Philosophy and Poetry clusters overlap in the high-rating, mid-price quadrant.
6. The correlation matrix confirms near-zero relationships between all numeric features.